

Delay-Doppler Domain Turbo Equalization for Single-Carrier Underwater Acoustic Communications

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I. INTRODUCTION

Abstract—Single carrier modulation, a commonly used method in high-speed underwater acoustic communications, requires equalization at the receiver. However, its performance remains sub-optimal in rapidly time-varying channels. Recently, delay-Doppler (DD) domain modulation, such as the well-known orthogonal time frequency space (OTFS) modulation, has garnered considerable attention for its ability to provide more reliable communication in highly dynamic environments. OTFS is essentially a precoded orthogonal frequency division multiplexing (OFDM) method that implements multicarrier modulation in the DD domain, yet it still has a high peak-to-average power ratio (PAPR). To maintain the low PAPR characteristic of single-carrier systems while enhancing performance in rapidly time-varying channels, this article proposes a DD domain turbo equalizer based on minimum mean square error (DD-MMSE-TEQ) for single-carrier underwater acoustic communications. The proposed equalizer comprises a DD domain equalizer and a time-domain decoder. Specifically, due to the inherent ability of DD domain transformation to convert time-varying channels into quasi-static channels, the proposed method demonstrates increased robustness to high Doppler spread. Moreover, through unitary transformation, the DD domain equalizer and time-domain decoder facilitate cross-domain soft information exchange, thereby reducing the probability of error propagation and enhancing the performance of iterative equalization. Simulation results indicate that the proposed DD-MMSE-TEQ outperforms frequency-domain turbo equalization at the cost of additional computational complexity at the receiver and achieves performance levels comparable to conventional OTFS in terms of bit error rate (BER). The proposed method has been validated using underwater experimental data collected from multiple experiments conducted in the Danjiangkou Reservoir, Henan Province, China, in July 2022. The results confirm that the proposed method surpasses other comparison methods in terms of BER.

Index Terms—Delay-Doppler (DD) domain, doubly selective channel, single-carrier underwater acoustic communications, turbo equalization.

Received 21 February 2024; revised 17 July 2024; accepted 22 October 2024. Date of publication 8 January 2025; date of current version 16 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62071383 and Grant 62471397, in part by the Fundamental Research Funds for the Central Universities under Grant 23GH02027, and in part by the Stable Supporting Fund of Acoustic Science and Technology Laboratory under Grant JCKYS2024604SSJS010. (Corresponding author: Chengbing He.)

Associate Editor: K. Pelekanakis.

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Digital Object Identifier 10.1109/JOE.2024.3508027

UNDERWATER acoustic (UWA) channels are one of the most complex doubly selective channels due to their unique characteristics [1]. For instance, strong background noise and significant transmission loss result in a low received signal-to-noise ratio (SNR). The limited available bandwidth restricts the data rate of UWA communications. The long delay spread introduces serious intersymbol interference (ISI). Additionally, the Doppler effect is more severe than in terrestrial wireless communications, resulting in rapidly time-varying channels. To mitigate ISI and the Doppler effect, many technologies have been studied in single-carrier (SC) UWA communications [2], [3], [4], [5], [6].

Single-carrier time-domain equalization (SC-TDE) has been successfully applied to UWA communications [2]. SC-TDE is able to provide better bandwidth efficiency because it does not require any guard intervals. A time-domain equalizer with iterative channel estimation was proposed, and the improved proportionate normalized least mean squares (IPNLMSs) algorithm was used for channel estimation [3]. However, due to insufficient observation data and unreliable symbol decisions, error propagation will be caused in the iterative channel estimation process. A low-complexity SC-TDE algorithm based on message passing was proposed to mitigate the error propagation [4]. Direct adaptive equalizer (DAE) does not require direct channel estimation but instead tracks channel dynamics by updating filter coefficients through adaptive algorithms such as LMS and RLS [5]. Existing sparse DAEs designed with proportional update or zero-attraction adaptive filtering principles were comprehensively compared in [5]. An iterative decision feedback equalization (DFE) performed both the feedforward and feedback filtering based on adaptive algorithms, which significantly reduced complexity and alleviated the slow-convergence problem commonly experienced by DAEs [6]. The DAE based on the minimum symbol error rate (MSER) criterion had been proven to be superior to the mean square error (MSE) criterion in terms of bit error rate (BER) [7]. The sparsity-aware equalizer was proposed for deep-sea vertical UWA communications, which utilizes the sparsity of UWA channels and MSER criteria to achieve error-free transmission [8]. However, SC-TDE is unstable and less robust due to the long delay spread and rapidly time-varying nature of the UWA channels.

Single-carrier frequency-domain equalization (SC-FDE) has garnered significant attention in UWA communications, which

can provide lower complexity and enhanced robustness in ISI and Doppler fading channels, and avoid issues such as high peak-to-average power ratio (PAPR) and sensitivity to frequency offset [9], [10]. Similar to orthogonal frequency division multiplexing (OFDM), SC-FDE also periodically inserts a cyclic prefix (CP) at the transmitter. In [11], a frequency-domain decision feedback equalizer (FD-DFE) based on iterative subblocks was proposed. The proposed method first compensated for the main inter-frequency interference (IFI) by time reversal processing and explicit IFI elimination, and then suppressed the residual IFI by FD-DFE. A block iterative decision feedback equalizer was proposed to achieve a performance and complexity tradeoff, with both feedforward and feedback filters being updated in the frequency-domain [12]. However, to design signals with zero padding and CP, SC-FDE typically requires information about the maximum multipath delay spread at the transmitter. Additionally, because the CP occupies a significant portion of the total frame, it reduces bandwidth efficiency.

Turbo equalization (TEQ) has emerged as a more promising equalization technology in SC UWA communications due to its outstanding error correction ability [13]. By iterating between the soft decoder and equalizer to exchange the soft information, detection reliability is significantly improved compared to traditional equalizers, which make hard decisions in a single iteration. Currently, turbo equalization is divided into a direct adaptive-based TEQ (DA-TEQ) [14], [15] and a channel estimation-based TEQ (CE-TEQ) [16], [17], [18], [19], depending on whether explicit channel estimation is required. A bi-directional time-domain DA-TEQ based on MSE criterion was proposed for single-carrier UWA communications, in which bidirectional gain was used to reduce the error propagation probability [14]. In [15], a frequency-domain DA-TEQ with a sliding window was proposed to accelerate the convergence of the filters. Although DA-TEQ adaptively updates the filter coefficients without channel estimation, it requires a longer training sequence to ensure that the filters converge to a steady state [16]. CE-TEQ calculates the filter coefficients according to the estimated channel, which has better robustness and performance despite its higher computational overhead. Qin et al. [16] proposed a turbo equalizer with Bayesian iterative channel estimation, reducing error propagation and significantly improving BER. In [17], a turbo equalizer with multitask sparse Bayesian channel estimation based on factor graphs was proposed for single-carrier UWA communications. A low-complexity frequency-domain CE-TEQ that integrates phase rotation compensation and soft successive interference cancellation was proposed [18]. The frequency-domain CE-TEQ method with iterative channel estimation was proposed for multiple-input-multiple-output (MIMO) UWA communications [19]. The methods described above perform well in both time-invariant and slowly time-varying channels. However, for rapidly time-varying UWA channels, the quasi-static assumption of the channels no longer holds true, resulting in frequency-domain turbo equalization performance degradation.

Recently, the orthogonal time-frequency space (OTFS) modulation was proposed to enhance the robustness of communication systems in high-speed mobile scenarios [20]. OTFS

transforms time-varying multipath channels into approximately time-invariant channels, and extends each symbol across the whole TF-domain, resulting in the same channel gain for all symbols, which greatly improves transmission reliability. Hence, OTFS modulation emerges as a viable alternative to conventional OFDM, particularly in rapidly time-varying channel scenarios. In [21], a low-complexity message-passing (MP) algorithm was proposed for joint interference cancellation and detection by exploiting the delay-Doppler (DD) domain channel sparsity. Xiang et al. [22] introduced a low-complexity Gaussian approximation MP (GAMP) OTFS detector, which further improved performance while maintaining low complexity. An iterative detection algorithm based on unitary approximate message passing (UAMP) was proposed in [23], which reduces computational complexity by leveraging the block circulant property of the OTFS channel matrix with ideal waveforms. In [24], a novel cross-domain iterative detection algorithm was introduced to enhance the performance of OTFS modulation in terms of error rates. Li and Yu [25] introduced a doubly iterative sparsified minimum MSE (DI-S-MMSE) turbo equalizer. To harness the inherent sparsity of the channel matrix, graph theory was applied to explore sparse matrices and formulated two sparsification criteria. Moreover, Jing et al. [26] proposed a novel 2-D passive time-reversal technique for OTFS UWA communications.

DD domain channel equalization was first introduced for UWA communications in [27]. More recently, orthogonal signal division multiplexing (OSDM) for UWA communications has also utilized DD domain processing [28], [29], [30]. OSDM proposed in [31] has been mathematically proven to be completely equivalent to OTFS [32]. The OSDM UWA communications system proposed in [28] utilized dedicated pilots for channel estimation, and tank test results indicated that OSDM outperforms traditional OFDM and SC systems with DFE. Based on the basis expansion model (BEM), a Doppler-resilient OSDM scheme was proposed to achieve reliable transmission in time-varying UWA channels [29]. In [30], the Doppler effect was modeled as a common time-varying phase, establishing the concept of intervector interference. An iterative per-vector equalization was proposed, integrating joint channel impulse response (CIR) and phase estimation, equalization, and decoding within a single loop. Although OTFS and OSDM partially address the issue of frequency offset sensitivity in multicarrier systems, the PAPR remains relatively high. This leads to power amplifiers operating in the nonlinear regime [33], [34].

Inspired by OTFS, this article proposes a DD domain equalizer based on MMSE for single-carrier UWA communications. The proposed equalizer enhances the performance of a single-carrier system in rapidly time-varying channels by transforming rapidly time-varying channels into quasi-static channels in the DD domain. Furthermore, the DD domain equalizer is extended to the coding system, and the DD domain turbo equalizer is proposed, namely, DD-MMSE-TEQ. The proposed method realizes cross-domain soft information interaction between the DD domain equalizer and time-domain decoder through unitary transformation, which reduces error propagation probability and improves the iteration gain. Both simulation and experimental results demonstrate the effectiveness of the proposed method.

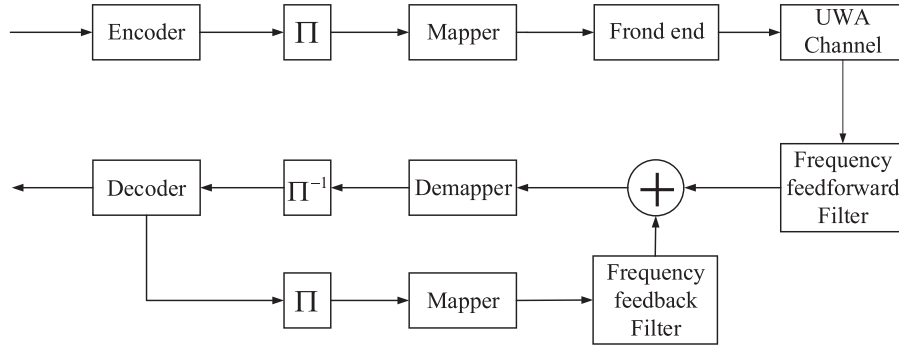


Fig. 1. Block diagram of frequency-domain turbo equalizer. Π and Π^{-1} are the interleaver and deinterleaver, respectively.

The proposed method does exceptionally well at handling time-varying channels and has several advantages, including lower PAPR and easier implementation.

The main contributions of this article are summarized as follows.

- 1) In this article, we formulate the input–output relationship of a single-carrier system in the DD domain. A novel equalizer operating in the DD domain is proposed for single-carrier UWA communications. The rapidly time-varying channels are transformed into DD domain quasi-static channels by the DD domain transform. Therefore, compared to equalizers operating in the TF-domain, the proposed equalizer operating in the DD domain improves the ability of single-carrier systems to process rapidly time-varying channels.
- 2) Additionally, by combining the proposed equalizer and turbo code, DD-MMSE-TEQ is proposed to further enhance the error performance. The equalizer operating in the DD domain and decoder operating in the time-domain realize cross-domain information exchange through unitary transformation, reducing the probability of error propagation and improving iterative gain.
- 3) Simulation results demonstrate that the proposed method outperforms frequency-domain turbo equalization in time-varying channels at the cost of increased receiver complexity. Additionally, the proposed method can achieve error performance comparable to traditional OTFS while significantly reducing PAPR compared to OTFS and OFDM.
- 4) Data obtained from a mobile UWA communications experiment conducted in July 2022 in the Danjiangkou Reservoir, Henan Province, China, further confirmed the effectiveness of the proposed method. The results demonstrate the exceptional performance of DD-MMSE-TEQ, particularly in communication scenarios characterized by significant delay spread and the Doppler effect. Almost error-free transmission was achieved at a communication distance of 2 km and a data rate of up to 4.076 kbps.

The rest of this article is organized as follows. In Section II, an overview of the single-carrier system model is provided. Section III introduces the single-carrier DD domain MMSE turbo equalization, including details about the DD domain channel models, DD domain equalization, and soft information interaction between the DD domain and time-domain.

The simulation results are presented in Section IV. Section V further validates the method's effectiveness using real-world UWA communications data. In Section VI, the computational complexity of the algorithms involved is analyzed. Finally, Section VII concludes this article.

II. SYSTEM MODEL

Fig. 1 shows a typical single-carrier transceiver with frequency-domain turbo equalization. The bitstream consists of encoded bits processed by a convolutional code with a rate of R_c . To mitigate consecutive burst errors during transmission and ensure accurate reconstruction at the receiver, a random interleaver is used to scramble the initial order of the encoded sequence. For 2^Q -ary modulation, given the constellation set $\mathcal{A} = [\alpha_1, \alpha_2, \dots, \alpha_{2^Q}]$, every Q bits $\mathbf{d}_k = [d_{k,1}, d_{k,2}, \dots, d_{k,Q}]$ are mapped to one symbol $x(k)$, where the corresponding bit pattern of α_i is $\mathbf{c}_i = \{c_{i,1}, c_{i,2}, \dots, c_{i,Q}\}$. After single-carrier modulation, the continuous baseband transmission signal is given by

$$s(t) = \sum_{k=0}^{K-1} x(k)g_{tx}(t - kT_s) \quad (1)$$

where $g_{tx}(t)$ is the transmitter shaping pulse, B is the system bandwidth, and $T_s = 1/B$ is the symbol duration.

The received signal after propagating through the time-varying UWA channels is expressed as

$$\begin{aligned} r(t) &= s(t) * h(\tau, t) + w(t) \\ &= \int_{-\infty}^{\infty} h(\tau, t)s(t - \tau)d\tau + w(t) \end{aligned} \quad (2)$$

where $w(t)$ is additive white Gaussian noise, with zero mean and variance σ^2 , denoted as $w \sim \mathcal{CN}(0, \sigma^2)$. The symbol $*$ represents linear equalization. The time-varying CIR is represented by $h(\tau, t)$. The discrete received signal is generated by match filtering $r(t)$ with $g_{rx}(t)$ and sampling at $t = kT_s$

$$r(k) = \sum_{l=0}^{L-1} h(k, l)x(k - l) + w(k) \quad (3)$$

where $w(k) = w(kT_s)$ is the discrete-time-domain noise and $g_{rx}(t)$ is the receiver shaping pulse.

The traditional single-carrier FD-DFE model is briefly reviewed before introducing the proposed method [12]. FD-DFE assumes that the channel remains static within a data block and converts the linear convolution to a circular convolution by adding a CP. As a result, in matrix form, (3) is rewritten

$$\mathbf{r} = \tilde{\mathbf{H}}\mathbf{x} + \mathbf{w} \quad (4)$$

where

$$\begin{aligned} \mathbf{r} &= [r(1), r(2), \dots, r(K)]^T \\ \mathbf{x} &= [x(1), x(2), \dots, x(K)]^T \end{aligned}$$

and

$$\mathbf{w} = [w(1), w(2), \dots, w(K)]^T$$

are the received symbols vector, transmitted symbols vector, and noise vector, respectively. $\tilde{\mathbf{H}}$ denotes the cyclic channel matrix

$$\tilde{\mathbf{H}} = \begin{bmatrix} h_0 & & h_{L-1} & \cdots & h_1 \\ h_1 & h_0 & & \ddots & \vdots \\ \vdots & h_1 & & h_{L-1} & \\ h_{L-1} & \vdots & h_0 & & \\ & h_{L-1} & h_1 & h_0 & \\ & & \vdots & \vdots & \ddots \\ & & h_{L-1} & h_{L-2} & \cdots & h_0 \end{bmatrix}. \quad (5)$$

The discrete Fourier transform (DFT) converts (4) into the frequency-domain

$$\begin{aligned} \mathbf{R} &= \mathbf{G}\mathbf{U} + \mathbf{W} \\ &= \mathbf{F}\tilde{\mathbf{H}}\mathbf{F}^H\mathbf{F}\mathbf{x} + \mathbf{F}\mathbf{w} \end{aligned} \quad (6)$$

where $\mathbf{G} = \mathbf{F}\tilde{\mathbf{H}}\mathbf{F}^H = \text{diag}\{G_0, G_1, \dots, G_{K-1}\}$ is a diagonal matrix, and its diagonal elements are the eigenvalues of $\tilde{\mathbf{H}}$. $\mathbf{U}$ and $\mathbf{W}$ are frequency-domain forms of the transmitted signal and noise, respectively. Using the FD-DFE scheme, the output of the equalizer at the i th iteration is written as [12]

$$\hat{\mathbf{x}}^{(i)} = \mathbf{F}^H \left(\mathbf{C}\mathbf{R} + \mathbf{B}\mathbf{F}\bar{\mathbf{x}}^{(i-1)} \right) \quad (7)$$

where $\bar{\mathbf{x}}^{(i-1)}$ is the soft decision symbol of the previous iteration detection. $\mathbf{C}$ and $\mathbf{B}$ are the frequency-domain feedforward filter coefficients and feedback filter coefficients, which are jointly optimized according to the MSE criterion.

The MSE is given by

$$J = \mathbb{E} \left[|\hat{x}^{(i)}(k) - x(k)|^2 \right] \quad (8)$$

where the expectation operator is denoted by $\mathbb{E}[\bullet]$. By applying Parseval's theorem, and using (7) and (8) can be rewritten as

$$J = \frac{1}{K^2} \sum_{k=0}^{K-1} \mathbb{E} \left[|C_k^{(i)} R_k + B_k^{(i)} \bar{U}_k^{(i-1)} - U_k|^2 \right]. \quad (9)$$

From (7) and by taking the expectation in (9) with respect to the data and noise, the MSE is expressed as

$$\begin{aligned} J &= \frac{1}{K^2} \sum_{k=0}^{K-1} |C_k^{(i)}|^2 \varrho_W^2 + |C_k^{(i)} G_k - 1|^2 \varrho_k^2 \\ &\quad + |B_k^{(i)}|^2 \varrho_k^{2,(i-1)} + 2\Re \left[B_k^{(i)*} (C_k^{(i)} G_k - 1) \rho_k^{(i-1)} \right] \end{aligned} \quad (10)$$

where $\varrho_k^2 = \mathbb{E}\{U_k^2\}$ and $\varrho_k^{2,(i-1)} = \mathbb{E}\{\bar{U}_k^{2,(i-1)}\}$ are the power of the transmission signal and previous iteration soft decision symbol in the frequency-domain. ϱ_W^2 is the noise power in the frequency-domain. The cross-correlation between the transmitted signal and the soft decision symbol is defined as $\rho_k^{(i-1)} = \mathbb{E}\{U_k (\bar{U}_k^{(i-1)})^*\}$. $\Re(x)$ is the real part of x . The gradient method is applied to minimize (10) with respect to the feedback filter coefficients under the constraint $\sum_{k=0}^{K-1} B_k^{(i)} = 0$. The Lagrange multiplier method is used to minimize the function

$$\begin{aligned} f(\mathbf{C}^{(i)}, \mathbf{B}^{(i)}, \lambda^{(i)}) &= \frac{1}{K^2} \sum_{k=0}^{K-1} |C_k^{(i)}|^2 \varrho_W^2 + |C_k^{(i)} G_k - 1|^2 \varrho_k^2 \\ &\quad + |B_k^{(i)}|^2 \varrho_k^{2,(i-1)} + 2\Re[B_k^{(i)*} (C_k^{(i)} G_k - 1) \rho_k^{(i-1)}] \\ &\quad + \lambda^{(i)} \sum_{k=0}^{K-1} B_k^{(i)} \end{aligned} \quad (11)$$

where $\lambda^{(i)}$ is the Lagrange multiplier.

Let us assume that the equalizer output signal $\hat{\mathbf{x}}^{(i)}$ and input signal $\mathbf{x}$ have the same statistical characteristics. Setting to zeros the gradient of (11) with respect to $C_k^{(i)}$, $B_k^{(i)}$, and $\lambda^{(i)}$, respectively, and the filter coefficients can be obtained as [12]

$$C_k^{(i)} = \frac{G_k^*}{\varrho_W^2 + \varrho_k^2 \left(1 - \frac{|\rho_k^{(i-1)}|^2}{\varrho_k^2 \varrho_k^{2,(i-1)}} \right) |G_k|^2}, \quad (12)$$

$$B_k^{(i)} = -\frac{\rho_k^{(i-1)}}{\varrho_k^{2,(i-1)}} \left(G_k C_k^{(i)} - \frac{1}{K} \sum_{k=0}^{K-1} G_k C_k^{(i)} \right). \quad (13)$$

In the initial iteration, no prior information is available, and the feedforward filter and feedback filter degenerate into

$$C_k^{(i)} = \frac{G_k^*}{\varrho_W^2 + |G_k|^2} \quad (14)$$

$$B_k^{(i)} = 0. \quad (15)$$

Consequently, FD-DFE degrades into a MMSE linear equalizer in the frequency-domain. Compared to a linear equalizer, FD-DFE can improve the detection performance of the current iteration by utilizing the prior information generated from the previous iteration. FD-DFE shows good performance in time-invariant or slowly time-varying channels because the assumption that the channels remain static in one block is established. However, in rapidly time-varying channels, this assumption is no longer satisfied [11].

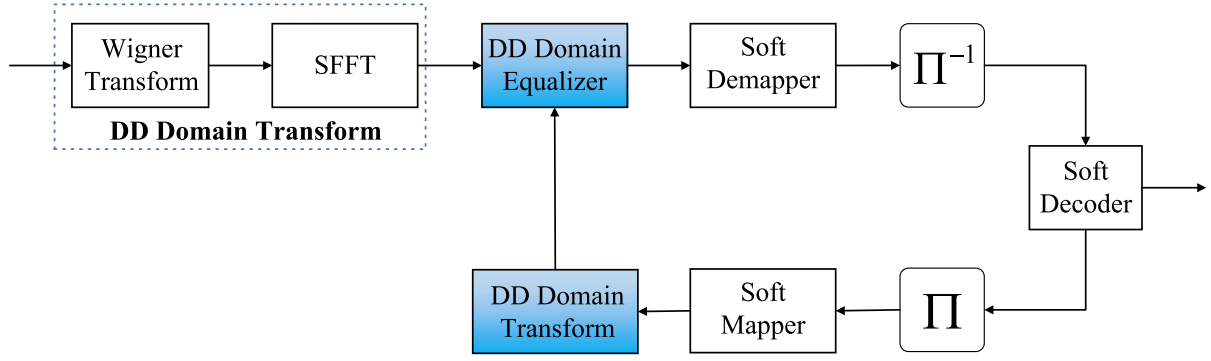


Fig. 2. Block diagram of SC-DD domain receiver, where Π and Π^{-1} are the interleaver and deinterleaver, respectively.

III. DD DOMAIN TURBO EQUALIZATION IN SINGLE-CARRIER UWA COMMUNICATION

Unlike a typical FD-DFE system, the proposed method incorporates a DD domain transform module at the receiver, which converts the time-domain signal into the DD domain for further processing. Fig. 2 shows the proposed single-carrier DD domain receiver. Because the equalizer operates in the DD domain, we formulate the input–output relationship of a single-carrier system in the DD domain.

A. DD Domain Channel

A doubly-selective fading DD domain channel with P paths is represented as

$$h(\tau, \nu) = \sum_{p=0}^{P-1} h_p \delta(\tau - \tau_p) \delta(\nu - \nu_p) \quad (16)$$

where h_p is the complex gain of the p th multipath, τ_p is the delay of the p th multipath, ν_p is the Doppler shift of the p th multipath, and δ represents the Dirac delta function. Given the system bandwidth B and frame length T_f , the delay and Doppler shift can be expressed as

$$\tau_p = \frac{l_p}{B}, \nu_p = \frac{\kappa_p}{T_f} \quad (17)$$

where $0 \leq l_p \leq L$ and $|\kappa_p| \leq \kappa_{\max}$ are integers, L is the maximum normalized delay, and $\kappa_{\max}$ denotes the maximum normalized Doppler.

B. Delay-Doppler Domain Input–Output Relation for Single-Carrier

According to the DD domain channel model, the received signal in (2) can be expressed as

$$r(t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\tau, \nu) s(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu + w(t). \quad (18)$$

Equation (18) demonstrates the received signal can be characterized as a weighted superposition of delayed and

frequency-shifted copies of the transmitted signal. The relationship between $h(\tau, t)$ and $h(\tau, \nu)$ is expressed as [35]

$$h(\tau, \nu) = \int_{-\infty}^{\infty} h(\tau, t) e^{-j2\pi\nu(t-\tau)} dt. \quad (19)$$

The above equation transforms the time-varying channels into approximately time-invariant channels in the DD domain. By substituting (16) and (17) into (18), the discrete form of (18) is reformulated as

$$\begin{aligned} r(k) &= \sum_{p=0}^{P-1} h_p e^{j2\pi \frac{\kappa_p(k-l_p)}{K}} x[k-l_p]_K + w(k) \\ &= \sum_{p=0}^{P-1} h_p \tilde{r}_p(k) + w(k) \end{aligned} \quad (20)$$

where $[\bullet]_K$ is the modulo- K operation and

$$\tilde{r}_p(k) \triangleq e^{j2\pi \frac{\kappa_p(k-l_p)}{K}} x[k-l_p]_K.$$

Then, the vector form of (20) is expressed by

$$\begin{aligned} \mathbf{r} &= \sum_{p=0}^{P-1} h_p \tilde{\mathbf{r}}_p(k) \\ &= \mathbf{H}_T \mathbf{x} + \mathbf{w} \end{aligned} \quad (21)$$

The time-domain channel matrix $\mathbf{H}_T$ is defined as

$$\mathbf{H}_T = \sum_{p=0}^{P-1} h_p \Pi^{l_p} \Delta^{\kappa_p}. \quad (22)$$

The effect of delay is represented by Π , which is a forward cyclic shift matrix and is labeled as

$$\Pi_{MN} = \begin{bmatrix} 0 & \cdots & 0 & 1 \\ 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix}_{MN \times MN} \quad (23)$$

where M and N are integers, and $K = M \times N$. Meanwhile, Δ represents the influence of Doppler, and it is an $MN \times MN$ diagonal matrix

$$\Delta = \text{diag}[\omega^0, \omega^1, \dots, \omega^{MN-1}], \quad \omega = e^{j\frac{2\pi}{MN}}. \quad (24)$$

For convenience, we will ignore the noise term and define $\mathcal{F}_{(M,N)}$ as the (M, N) point DD domain transform. The DD domain received symbols matrix is obtained as

$$\begin{aligned} \mathbf{Y}_{\text{DD}} &= \mathcal{F}_{(M,N)}\{\mathbf{r}\} \\ &= \sum_{p=0}^{P-1} h_p \mathcal{F}_{(M,N)}\{\tilde{\mathbf{r}}_p\}. \end{aligned} \quad (25)$$

The virtual DD domain transmitted symbols matrix is defined as $\mathbf{X}_{\text{DD}} = \mathcal{F}_{(M,N)}\{\mathbf{x}\}$. The DD domain form of the received symbols vector $\tilde{\mathbf{r}}_p$ corresponding to the p th path is given by [36]

$$\begin{aligned} \mathcal{F}_{(M,N)}\{\tilde{\mathbf{r}}_p\}[m, n] &= e^{j2\pi\kappa_p \frac{[m-l_p]M}{K}} \mathcal{F}_{(M,N)}\{\mathbf{x}\}[m-l_p, n-\kappa_p] \\ &= e^{j2\pi\kappa_p \frac{[m-l_p]M}{K}} \\ &\begin{cases} \mathbf{X}_{\text{DD}}\{[m-l_p]_M, [n-\kappa_p]_N\}, & m \geq l_p \\ \mathbf{X}_{\text{DD}}\{[m-l_p]_M, [n-\kappa_p]_N\}e^{-j2\pi \frac{n}{N}}, & m < l_p \end{cases} \end{aligned} \quad (26)$$

where $m \in [0, M-1]$, $n \in [0, N-1]$. According to (25) and (26), the single-carrier DD domain input-output relationship is given by

$$\mathbf{Y}_{\text{DD}}[m, n] = \begin{cases} \sum_{p=0}^{P-1} h_p e^{j2\pi\kappa_p \frac{[m-l_p]M}{K}} \mathbf{X}_{\text{DD}}\{[m-l_p]_M, [n-\kappa_p]_N\}, & m \geq l_p \\ \sum_{p=0}^{P-1} h_p e^{j2\pi(\kappa_p \frac{[m-l_p]M}{K} - \frac{n}{N})} \mathbf{X}_{\text{DD}}\{[m-l_p]_M, [n-\kappa_p]_N\}, & m < l_p. \end{cases} \quad (27)$$

Equation (27) shows that the received signal in the DD domain can be characterized as a linear combination of $(m-l_p, n-\kappa_p)$ th received symbol in the identical domain for a given pair of delay tap l_p and Doppler tap κ_p .

C. DD-MMSE-TEQ

The received signal $r(k)$ is initially rearranged columnwise into a matrix $\mathbf{Y}_{\text{T}}$ with a size of $M \times N$. Considering a rectangular received pulse shaping waveform, M -point DFT is applied to each column of $\mathbf{Y}_{\text{T}}$, resulting in the TF-domain received signal $\mathbf{Y}_{\text{TF}}$ as

$$\mathbf{Y}_{\text{TF}} = \mathbf{F}_M \mathbf{G}_{\text{rx}} \mathbf{Y}_{\text{T}} = \mathbf{F}_M \mathbf{I}_M \mathbf{Y}_{\text{T}} \quad (28)$$

where $\mathbf{F}_M$ is the M -point Fourier transform matrix, $\mathbf{G}_{\text{rx}}$ is the rectangular window, and $\mathbf{I}_M$ is the $M \times M$ identity matrix. The above equation is referred to as the Wigner transform [37]. Subsequently, the TF-domain signal $\mathbf{Y}_{\text{TF}}$ undergoes a symplectic fast Fourier transform (SFFT), which entails performing M -point IDFT transformation columnwise and N -point DFT transformation rowwise. The DD domain received signal $\mathbf{Y}_{\text{DD}}$ is obtained as

$$\mathbf{Y}_{\text{DD}} = \mathbf{F}_M^H \mathbf{Y}_{\text{TF}} \mathbf{F}_N = \mathbf{I}_M \mathbf{Y}_{\text{T}} \mathbf{F}_N. \quad (29)$$

Since equalization is performed in the DD domain, we directly provide the relationship between the DD domain channel matrix

and time-domain channel matrix as follows [24]:

$$\mathbf{H}_{\text{DD}} = (\mathbf{F}_N \otimes \mathbf{I}_M) \mathbf{H}_{\text{T}} (\mathbf{F}_N^H \otimes \mathbf{I}_M) \quad (30)$$

where $\otimes$ represents the Kronecker product, and $(\bullet)^H$ denotes the Hermitian transpose.

According to the MMSE criterion, to recover the transmitted signal from the distorted signal, the goal is to minimize the l_2 norm between the transmitted symbol and equalized symbol to obtain the optimal filter coefficients. In other words, we need to find the equalizer coefficients that minimize the MSE $\mathbb{E}\{\|\mathbf{x}_{\text{T}} - \hat{\mathbf{x}}_{\text{T}}\|^2\}$, which refers to the difference between the transmitted and equalized symbol. This problem can be modeled as [38]

$$\hat{\mathbf{x}}_{\text{DD}}(k) = \bar{\mathbf{x}}_{\text{DD}}(k) + \mathbf{C}_{\bar{\mathbf{x}}_{\text{DD}}(k), \mathbf{y}_{\text{DD}}} \mathbf{C}_{\mathbf{y}_{\text{DD}}, \mathbf{y}_{\text{DD}}}^{-1} (\mathbf{y}_{\text{DD}} - \mathbb{E}\{\mathbf{y}_{\text{DD}}\}) \quad (31)$$

where $\hat{\mathbf{x}}_{\text{DD}}(k)$ represents the estimated value of the transmitted signal in the DD domain, and $\bar{\mathbf{x}}_{\text{DD}}(k)$ is the mean of the transmitted signal in the DD domain.

$$\mathbf{y}_{\text{DD}} = \text{vec}(\mathbf{Y}_{\text{DD}}) = (\mathbf{F}_N \otimes \mathbf{I}_M) \mathbf{y}_{\text{T}}$$

is the vector of the received signal in the DD domain. The covariance matrix between the vectors is represented as $\mathbf{C}_{\mathbf{a}, \mathbf{b}} = \text{Cov}\{\mathbf{a}, \mathbf{b}\} = \mathbb{E}\{\mathbf{a}\mathbf{b}^H\} - \mathbb{E}\{\mathbf{a}\}\mathbb{E}\{\mathbf{b}^H\}$. Assuming that the transmitted signal is uncorrelated, which holds approximately due to the interleaver. According to the covariance formula, the covariance matrices for the transmitted signal, received signal, and noise can be obtained

$$\mathbf{V} = \text{Cov}\{\mathbf{x}_{\text{T}}, \mathbf{x}_{\text{T}}\} = \text{diag}[v_1, \dots, v_K] \quad (32a)$$

$$\mathbf{\Psi} = \text{Cov}\{\mathbf{y}_{\text{DD}}, \mathbf{y}_{\text{DD}}\} = \mathbf{H}_{\text{DD}} \mathbf{V} \mathbf{H}_{\text{DD}}^H + \sigma \mathbf{I} \quad (32b)$$

$$\begin{aligned} \text{Cov}\{\mathbf{w}_{\text{DD}}, \mathbf{w}_{\text{DD}}\} &= (\mathbf{F}_N \otimes \mathbf{I}_M) \text{Cov}\{\mathbf{w}_{\text{T}}, \mathbf{w}_{\text{T}}\} (\mathbf{F}_N^H \otimes \mathbf{I}_M) \\ &= \sigma \mathbf{I}_{MN} \end{aligned} \quad (32c)$$

$$\text{Cov}\{\mathbf{x}_{\text{DD}}(k), \mathbf{y}_{\text{DD}}\} = \text{Cov}\{\mathbf{x}_{\text{DD}}(k), \mathbf{x}_{\text{DD}}\} \mathbf{H}_{\text{DD}}^H = v_k \mathbf{h}_k^H. \quad (32d)$$

Substituting (32) into (31), the equalized symbol in the DD domain can be expressed as

$$\begin{aligned} \hat{\mathbf{x}}_{\text{DD}}(k) &= \bar{\mathbf{x}}_{\text{DD}}(k) + v_k \mathbf{h}_k^H \mathbf{\Psi}^{-1} (\mathbf{y}_{\text{DD}} - \mathbf{H}_{\text{DD}} \bar{\mathbf{x}}_{\text{DD}}) \\ &= \bar{\mathbf{x}}_{\text{DD}}(k) + v_k \mathbf{f}_k^H (\mathbf{y}_{\text{DD}} - \mathbf{H}_{\text{DD}} \bar{\mathbf{x}}_{\text{DD}}) \end{aligned} \quad (33)$$

where $\mathbf{f}_k^H = \mathbf{h}_k^H \mathbf{\Psi}^{-1}$, and $\mathbf{h}_k$ represents the k th column of $\mathbf{H}_{\text{DD}}$.

However, the derivation of $\hat{\mathbf{x}}_{\text{DD}}(k)$ is related to the priori log-likelihood ratio (LLR) $L(d_{k,j})$, causing the extrinsic LLR $L_e(d_{k,j})$ obtained through $\hat{\mathbf{x}}_{\text{DD}}(k)$ to be correlated with the a priori LLR $L(d_{k,j})$. By setting the mean $\bar{\mathbf{x}}_{\text{DD}}(k) = 0$ and variance $v_k = 1$ to eliminate its own a priori, the equalized symbol is reexpressed as [39]

$$\begin{aligned} \hat{\mathbf{x}}_{\text{DD}}(k) &= 0 + 1 \cdot \mathbf{h}_k^H (\mathbf{\Psi} + (1 - v_k) \mathbf{h}_k \mathbf{h}_k^H)^{-1} (\mathbf{y} - \mathbf{H}_{\text{DD}} \bar{\mathbf{x}}_{\text{DD}} \\ &\quad + \bar{\mathbf{x}}_{\text{DD}}(k) \mathbf{h}_k) \\ &= 0 + 1 \cdot \mathbf{f}_k^H (\mathbf{y} - \mathbf{H}_{\text{DD}} \bar{\mathbf{x}}_{\text{DD}} + \bar{\mathbf{x}}_{\text{DD}}(k) \mathbf{h}_k). \end{aligned} \quad (34)$$

To avoid repeated matrix inversion in the calculation of $\hat{x}_{DD}(k)$, the Woodbury matrix identity is creatively proposed in [40]. The scaled version of $\mathbf{f}_k$ can be obtained as

$$\begin{aligned}\hat{\mathbf{f}}_k &\triangleq (\mathbf{\Psi} + (1 - v_k)\mathbf{h}_k\mathbf{h}_k^H)^{-1}\mathbf{h}_k \\ &\triangleq (\mathbf{\Psi}^{-1} - \mathbf{\Psi}^{-1}\mathbf{h}_k((1 - v_k)^{-1} + \mathbf{h}_k^H\mathbf{\Psi}^{-1}\mathbf{h}_k)^{-1}\mathbf{h}_k^H\mathbf{\Psi}^{-1})\mathbf{h}_k \\ &= \mathbf{f}_k - \mathbf{f}_k((1 - v_k)^{-1} + \mathbf{f}_k^H\mathbf{h}_k)^{-1}\mathbf{f}_k^H\mathbf{h}_k \\ &= (1 + (1 - v_k)\mathbf{f}_k^H\mathbf{h}_k)^{-1}\mathbf{f}_k \\ &= \beta_k\mathbf{f}_k\end{aligned}\quad (35)$$

where $\beta_k = (1 + (1 - v_k)\mathbf{f}_k^H\mathbf{h}_k)^{-1}$ is a scaled factor. Substituting (35) into (34), the DD domain estimated symbol is calculated as

$$\begin{aligned}\hat{x}_{DD}(k) &= \beta_k\mathbf{f}_k^H(\mathbf{y}_{DD} - \mathbf{H}_{DD}\bar{\mathbf{x}}_{DD} + \bar{x}_{DD}(k)\mathbf{h}_k) \\ &= \frac{\mathbf{h}_k^H\mathbf{\Psi}^{-1}(\mathbf{y}_{DD} - \mathbf{H}_{DD}\bar{\mathbf{x}}_{DD}) + \bar{x}_{DD}(k)\xi_k}{1 + (1 - v_k)\xi_k}\end{aligned}\quad (36)$$

where $\xi_k = \mathbf{f}_k^H\mathbf{h}_k$ is a real number due to $\mathbf{\Psi}$ is a Hermitian matrix. It's worth mentioning that, during the first iteration, $\bar{\mathbf{x}}_{DD}$ and v_k are initialized as $\bar{\mathbf{x}}_{DD} = \mathbf{0}$ and $v_k = 1$. In the subsequent iterations, they are updated by the soft decoder. Because modulation is not performed in the DD domain, the equalized symbols in the DD domain do not contain any statistical information. To calculate soft information, the equalized symbols in the DD domain are converted back to the time-domain and represented as

$$\hat{\mathbf{X}}_T = \mathbf{G}_{tx}\mathbf{F}_M^H\hat{\mathbf{X}}_{TF} = \mathbf{G}_{tx}\hat{\mathbf{X}}_{DD}\mathbf{F}_N^H\quad (37)$$

where $\hat{\mathbf{X}}_{DD} = \text{vec}^{-1}(\hat{\mathbf{x}}_{DD})$, $\hat{\mathbf{X}}_{TF} = \mathbf{F}_M\hat{\mathbf{X}}_{DD}\mathbf{F}_N^H$.

After obtaining the estimated time-domain signal, the estimated symbol $\hat{x}_T(k)$ is mapped to its corresponding LLR for soft decoding. Subsequent soft information calculations are all performed in the time-domain. For convenience, we omit the subscript $(\bullet)_T$. The posterior probability $P(d_{k,j}|\mathbf{y})$ of each bit relative to the entire received sequence needs to be derived. However, computing $P(d_{k,j}|\mathbf{y})$ is very time-consuming, especially when K is large. In [39], a creative approach is introduced to approximate $P(d_{k,j}|\mathbf{y})$ by utilizing the posterior probability regarding the estimated symbol $P(d_{k,j}|\hat{x}(k))$. The LLR for $P(d_{k,j}|\hat{x}(k))$ is defined as

$$\begin{aligned}L(d_{k,j}|\hat{x}(k)) &= \ln \frac{P(d_{k,j} = 0|\hat{x}(k))}{P(d_{k,j} = 1|\hat{x}(k))} \\ &= \ln \frac{\sum_{\forall \mathbf{d}_k: d_{k,j}=0} p(\hat{x}(k)|\mathbf{d}_k)P(\mathbf{d}_k)}{\sum_{\forall \mathbf{d}_k: d_{k,j}=1} p(\hat{x}(k)|\mathbf{d}_k)P(\mathbf{d}_k)} \\ &= \ln \frac{\sum_{\forall \mathbf{d}_k: d_{k,j}=0} p(\hat{x}(k)|\mathbf{d}_k) \prod_{\forall j': j' \neq j} P(d_{k,j'})}{\sum_{\forall \mathbf{d}_k: d_{k,j}=1} p(\hat{x}(k)|\mathbf{d}_k) \prod_{\forall j': j' \neq j} P(d_{k,j'})} \\ &\quad + \ln \frac{P(d_{k,j} = 0)}{P(d_{k,j} = 1)}.\end{aligned}\quad (38)$$

The first term in (38) is defined as the extrinsic LLR $L_e(d_{k,j})$, and the second term represents the priori LLR $L(d_{k,j})$ derived from the decoder. It is observed that the extrinsic LLR $L_e(d_{k,j})$ is obtained by subtracting the priori LLR $L(d_{k,j})$ from the

posterior LLR $L(d_{k,j}|\hat{x}(k))$, where $L_e(d_{k,j})$ and $L(d_{k,j})$ are strictly independent.

From the observation in (38), it is noted that the $L_e(d_{k,j})$ calculation depends on two parameters, $L(d_{k,j}|\hat{x}(k))$ and $L(d_{k,j})$. The computation of $L(d_{k,j}|\hat{x}(k))$ relies on a statistical model. Assuming that the equalized symbol follows a Gaussian distribution as

$$p(\hat{x}(k) | x(k) = \alpha_i) = \frac{1}{\pi\varsigma_k^2} \exp\left(-\frac{|\hat{x}(k) - \mu_k\alpha_i|^2}{\varsigma_k^2}\right)\quad (39)$$

where μ_k and ς_k^2 represent the mean and variance of the estimation error $e(k) = \hat{x}(k) - \alpha_i$, and are expressed as

$$\mu_k = \mathbb{E}\{\hat{x}(k)|x(k) = \alpha_i\}\quad (40)$$

$$\varsigma_k^2 = \mathbb{E}\{|\hat{x}(k) - \mu_k|^2|x(k) = \alpha_i\}.\quad (41)$$

Based on the estimated symbol, the mean and variance can be further expressed as

$$\mu_k = \frac{\xi_k\alpha_i}{1 + (1 - v_k)\xi_k}\quad (42)$$

$$\varsigma_k^2 = \frac{\xi_k(1 - v_k\xi_k)}{[1 + (1 - v_k)\xi_k]^2}.\quad (43)$$

Substituting the mean and variance into (38), we obtain the extrinsic LLR for the equalizer as

$$\begin{aligned}L_e(d_{k,1}) &= 2\sqrt{2}\text{Re}\{\hat{x}(k)\}/(1 - \mu_k) \\ &= \frac{2\sqrt{2}[1 + (1 - v_k)\xi_k]\text{Re}\{\hat{x}(k)\}}{1 - v_k\xi_k}\end{aligned}\quad (44a)$$

$$\begin{aligned}L_e(d_{k,2}) &= 2\sqrt{2}\text{Im}\{\hat{x}(k)\}/(1 - \mu_k) \\ &= \frac{2\sqrt{2}[1 + (1 - v_k)\xi_k]\text{Im}\{\hat{x}(k)\}}{1 - v_k\xi_k}.\end{aligned}\quad (44b)$$

The priori LLR is denoted as

$$P(d(k,j) = c_{i,j}) = \frac{1}{2} \left[1 + \tilde{c}_{i,j} \tanh\left(\frac{L_a^D(d(k,j))}{2}\right) \right]\quad (45)$$

where

$$\tilde{c}_{i,j} = \begin{cases} +1, & c_{i,j} = 0 \\ -1, & c_{i,j} = 1. \end{cases}\quad (46)$$

By combining (38) through (46), the extrinsic LLR of the equalized symbol is calculated. After deinterleaving, the extrinsic LLR is fed into the decoder as the a priori LLR. The decoder will output two values, namely, the estimated information bits and the posterior LLR. Combining the priori probability in (45), the priori mean and variance of $\hat{x}(k)$ can be calculated as

$$\bar{x}(k) = \sum_{\alpha_i \in \mathcal{A}} \alpha_i \prod_{j=1}^Q P(d(k,j) = c_{i,j})\quad (47)$$

$$v_k = \sum_{i=1}^{2^Q} |x(k) - \bar{x}(k)|^2 P(x(k) = \alpha_i) = 1 - |\bar{x}(k)|^2.\quad (48)$$

The priori mean $\bar{x}(k)$ needs to be transformed into the DD domain before being fed into the DD domain equalizer. Similar

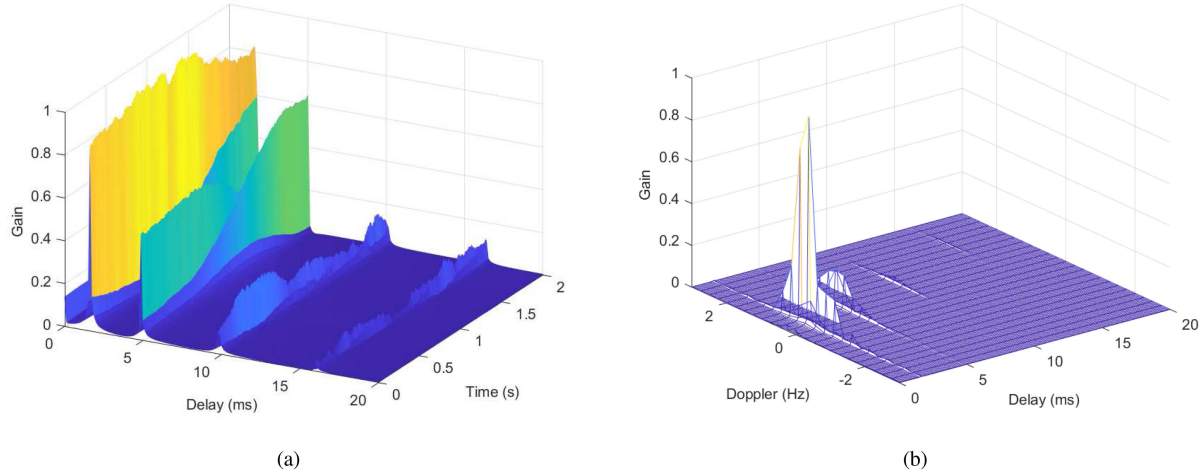


Fig. 3. Doubly selective UWA channel for simulation. (a) Channel in time-domain. (b) Channel in DD domain.

Algorithm 1: Proposed Channel Equalization Algorithm (DD-MMSE-TEQ).

Initialize: The prior mean $\bar{x}(k) = 0$ and variance $v_k = 1$;
Input: received signal r , noise power σ^2 , number of iterations $I_{\max}$, block length K ;

- 1: **for** each $i \in [1, I_{\max}]$ **do**
- 2: perform (22) to obtain $\mathbf{H}_T$;
- 3: perform (28) and (29) to obtain $\mathbf{Y}_{DD}$;
- 4: perform (30) to obtain $\mathbf{H}_{DD}$;
- 5: **for** each $k \in [0, K - 1]$ **do**
- 6: perform (36) to obtain $\hat{x}_{DD}(k)$;
- 7: **end for**
- 8: perform (37) to obtain $\hat{\mathbf{X}}_T$;
- 9: update $\bar{x}(k)$ and v_k from (47) and (48);
- 10: perform (49) to update $\bar{\mathbf{X}}_{DD}$;
- 11: **end for**
- 12: **return** estimated bit sequence $\hat{\mathbf{d}}$.

to (29), the priori mean in the DD domain is represented as

$$\bar{\mathbf{X}}_{DD} = \mathbf{F}_M^H \bar{\mathbf{X}}_{TF} \mathbf{F}_N = \mathbf{I}_M \bar{\mathbf{X}}_T \mathbf{F}_N. \quad (49)$$

It should be noted that the prior mean and variance are set to 0 and 1 in the first iteration, and they are updated by the decoder as the iteration progresses. The proposed algorithm is summarized in Algorithm 1.

IV. SIMULATION RESULTS

In this section, we utilize simulations to assess the performance of the proposed method. The UWA channel is generated according to the statistical channel model proposed in [41]. The communication configuration includes a 5-km transmission distance, a carrier frequency of 12 kHz, and a bandwidth of 6 kHz. The water depth is 50 m, with the transmitter and receiver approximately 25 m below the water surface. The relative speed of the transceiver is set at 10 knots. The channel coding employs a recursive convolutional (RSC) code with a rate of 1/2 and a

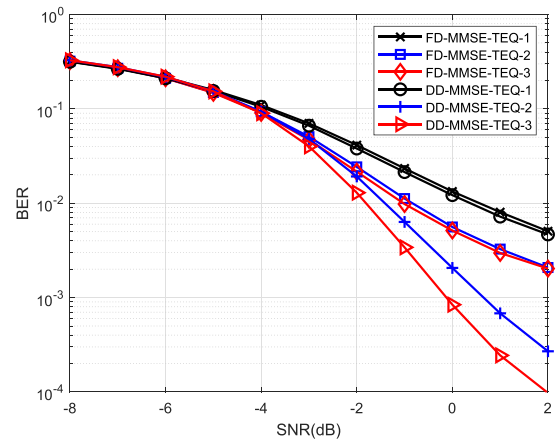


Fig. 4. BER performance with different turbo iterations and methods.

generator polynomial $[G1, G2] = [5, 7]$. Quadrature phase shift keying (QPSK) modulation is employed.

The impulse response of the time-domain channel is shown in Fig. 3(a), while the DD domain channel is presented in Fig. 3(b). The maximum delay spread of the channel is approximately 10 ms, which is roughly equivalent to 80 symbol intervals, with a Doppler frequency shift of about 1.5 Hz. Comparing the time-domain channel and the DD domain channel, it is found that there are significant differences between the time-domain channel and the DD domain channel. The intricately time-varying multipath channel in the time-domain shows clear and simple structural characteristics in the DD domain.

Fig. 4 illustrates the BER performance comparison between the proposed single-carrier DD domain turbo equalization (DD-MMSE-TEQ) and single-carrier frequency-domain turbo equalization (FD-MMSE-TEQ) [19]. A block length $K = M \times N$, which is set to 1024, and a pilot sequence of 256 symbols is employed for channel estimation. Analyzing Fig. 4, it is obvious that DD-MMSE-TEQ greatly outperforms FD-MMSE-TEQ, owing to its efficient utilization of the DD domain channel characteristics. It delivers a performance boost of approximately one

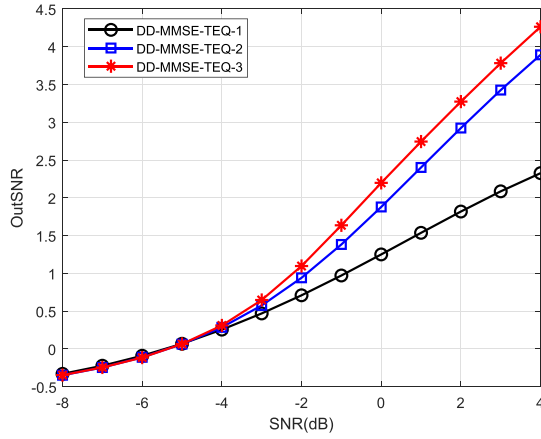


Fig. 5. OSNR of the DD-MMSE-TEQ.

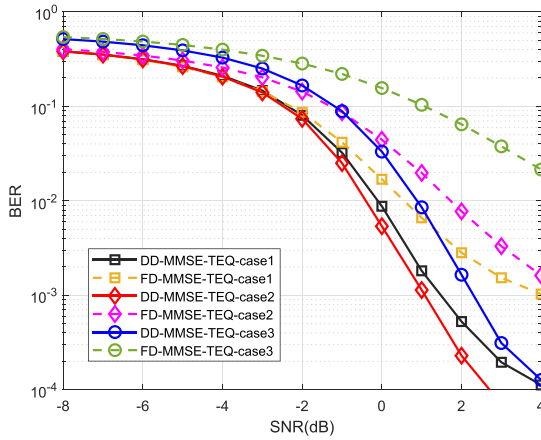


Fig. 6. Performance of the proposed method under different block length. Case1: $K=1024$, case2: $K=2048$, case3: $K=4096$.

order of magnitude after three iterations at an SNR of 2 dB. Furthermore, DD-MMSE-TEQ achieves an iteration gain of roughly 4 dB. Compared to FD-MMSE-TEQ, the iterative equalization performance is significantly improved. The improvement can be largely attributed to the cross-domain information exchange between the time-domain decoder and DD domain equalizer. After equalization in the DD domain, the continuous burst errors in the time-domain are mitigated, reducing the probability of error propagation.

The output SNR (OSNR) of the equalized symbol is defined as

$$\gamma = 10 \log_{10} \frac{\|x(k)\|_2^2}{\|\hat{x}(k) - x(k)\|_2^2}. \quad (50)$$

The OSNR of the proposed method, illustrated in Fig. 5, exhibits a gradual increase as the iterations progress. The OSNR gain of roughly 2 dB is realized after three iterations, further confirming the validity of the proposed method.

Fig. 6 shows the BER performance for various block durations, including $K = 1024$ (case1), $K = 2048$ (case2) and $K = 4096$ (case3). The performance of DD-MMSE-TEQ improves as frame length increases because a larger block can decrease the fractional Doppler and fractional delay effects.

However, the performance will not improve as the block length increases blindly. On the contrary, overlong block length can lead to performance degradation because the quasi-static channel assumption no longer holds strictly. Additionally, overlong block length results in unacceptable computational overhead. On the contrary, the performance of FD-MMSE-TEQ deteriorates due to the larger block durations, which further undermine the assumption that the channel is quasi-static within one block.

Fig. 7(a) shows a comparison between the uncoded and coded BER performance of DD-MMSE-TEQ and that of conventional OTFS. There is still a slight BER and symbol error rate (SER) performance gap between the proposed method and conventional OTFS, particularly in the high SNR range. Nevertheless, the slightly inferior BER performance gap can be almost ignored. Fig. 7(b) illustrates the PAPR characteristics of SC, OTFS, and OFDM signals in terms of the complementary cumulative distribution function (CCDF). Although the PAPR of OTFS is proportional to the number of Doppler bins N and is lower than that of OFDM. However, SC achieves significantly lower PAPR compared to OTFS and OFDM. Therefore, DD-MMSE-TEQ is an effective method in view of the characteristics of PAPR.

V. EXPERIMENTAL RESULTS

A. Experiment Description

We conducted an experiment at the Danjiangkou Reservoir in Henan Province in July 2022, focusing on single-carrier mobile UWA communications. The receiving array consisted of eight vertically placed hydrophones, each spaced 0.25 m apart. The communication distance was 2 km. The average water depth was 50 m in the experimental waters, with the top receiving hydrophone positioned approximately 20 m below the water surface. When the ship was stationary, the transmitter was located about 20 m below the water surface. When the ship was moving toward or away from the receiving hydrophone at speeds ranging from 2 to 4 knots, the transmitter depth varied between 3 and 7 m below the water surface. The carrier frequency was 16 kHz, the sampling frequency was 48 kHz, and the bandwidth was 6 kHz. The experiment scenario is shown in Fig. 8. Table I summarizes the main parameters of the mobile UWA communications experiment.

The frame structure is shown in Fig. 9. During the experiment, a total of 138 240 bits were divided into 15 frames, with each frame including 12 data blocks and a pilot block composed entirely of pseudonoise (PN) sequence, totaling 13 568 QPSK symbols. Each data block consisted of 768 QPSK symbols and 256 PN symbols. The source was encoded using RSC code with a code rate of 1/2 and the generator polynomial $[G1, G2] = [5, 7]$. The baseband signal was added with a synchronization signal and a guard interval, then it was modulated by a carrier to form a passband transmission signal. The 2-segment linear frequency modulation (LFM) signals, with a bandwidth of 6 kHz and a duration of 0.1 s, were used for synchronization, coarse Doppler estimation, and compensation. The IDSig was used as an identification signal for easy identification. A guard interval of 0.1 s was added between the LFM signal and payload sequence to prevent interference. On the receiver, the received signals were

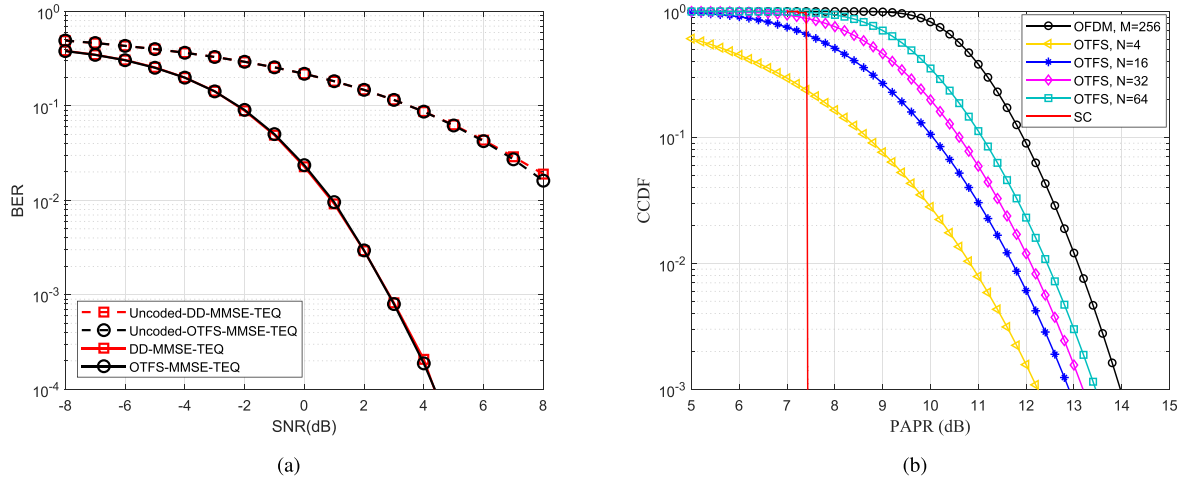


Fig. 7. Performance comparison between OTFS and DD-MMSE-TEQ. (a) BER and SER of DD-MMSE-TEQ compared to OTFS. (b) PAPR of SC compared to OFDM and OTFS.

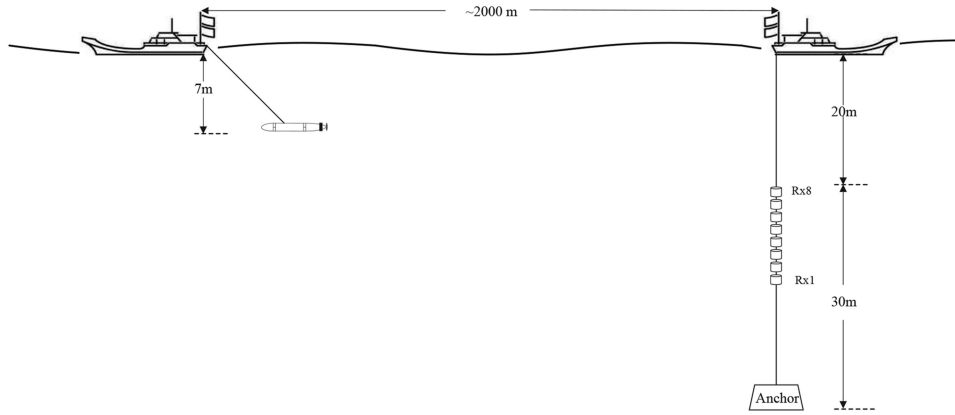


Fig. 8. Deployment of the experiment.

TABLE I
PARAMETER SETTINGS OF EXPERIMENT

Parameters	Value	Parameters	Value
Coding scheme	RSC	Sampling rate	48 kHz
Generator polynomial	[5,7]	Water depth	50 m
Code rate	1/2	Communication distance	2 km
Modulation order	QPSK	Receiver depth	20 m
Carrier frequency	16 kHz	Transmitter depth	20 m(3–7 m)
Bandwidth	6 kHz	Array element interval	0.25 m
Number of receiving hydrophones	8	Relative speed	2–4 knots

demodulated, lowpass filtered, and downsampled to form the baseband signals after synchronization, Doppler estimation and compensation. The transmission time for one frame, T_f , was 2.2613 s, resulting in a data rate of approximately

$$R = \frac{N_b}{T_f} = \frac{(N_t - N_p)R_c \log_2 M}{T_f} = 4.076 \text{ kbps} \quad (51)$$

where N_b , N_t , and N_p are the number of source bits, QPSK symbols, and pilot symbols, respectively. The time-varying CIRs estimated from the data collected by the top hydrophone during

the experiment are shown in Fig. 10(a)–(c), respectively. The channel spreading functions (SF) in the DD domain are shown in Fig. 10(d)–(f), respectively. To illustrate the multipath structure more clearly, the instantaneous CIRs of the first PN block are shown in Fig. 11.

Due to the shallow water in the experiment area, the multipath spread was significant regardless of whether the transceiver was stationary or moving. The channel energy was mainly concentrated on a few taps, indicating that the channel was sparse. During the DJKC1R2 experiment, where the transceiver was static, the Doppler shift of the channel is only 0.5 Hz, but

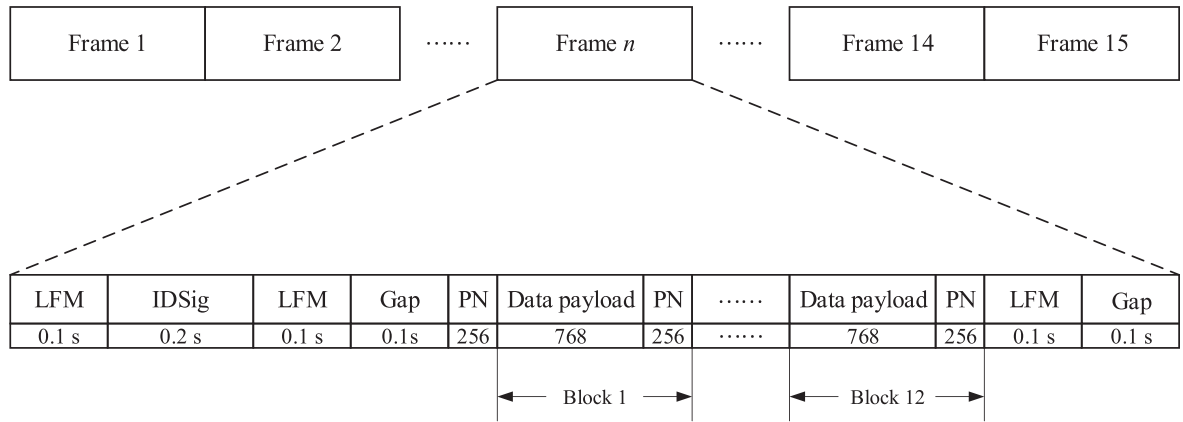


Fig. 9. Frame structure of the transmit signal in the lake experiment.

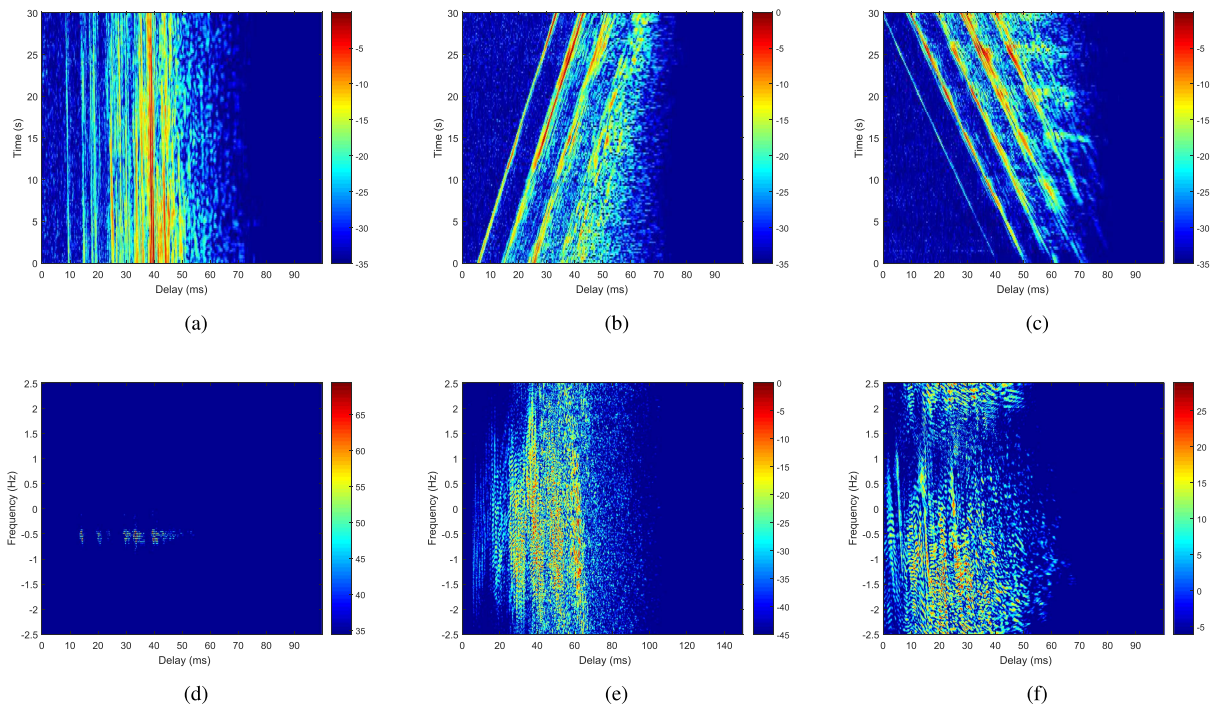


Fig. 10. Estimated UWA channels from real-world experiments. (a) CIR without Doppler compensation for DJKC1R2. (b) CIR without Doppler compensation for DJKE1R1. (c) CIR without Doppler compensation for DJKG1R2. (d) SF function for DJKC1R2. (e) SF function for DJKE1R1. (f) SF function for DJKG1R2.

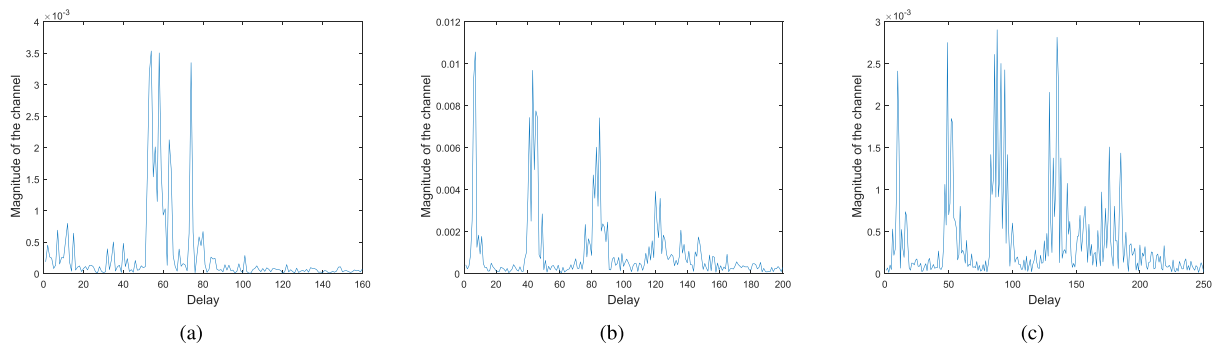


Fig. 11. Estimated instantaneous CIRs from experiments. (a) DJKC1R2. (b) DJKE1R1. (c) DJKG1R2.

TABLE II
AVERAGE BER PERFORMANCE OF EXPERIMENT DJKC1R2 WITH DIFFERENT METHODS

Channel Number	Method	Iter=1	Iter=2	Iter=3
1	TDDA-TEQ	4.39×10^{-1}	4.36×10^{-1}	4.39×10^{-1}
	TDMSER-TEQ	4.38×10^{-1}	4.35×10^{-1}	4.34×10^{-1}
	FD-MMSE-TEQ	1.97×10^{-1}	2.00×10^{-1}	1.89×10^{-1}
	DD-MMSE-TEQ	1.45×10^{-1}	9.22×10^{-2}	7.51×10^{-2}
2	TDDA-TEQ	3.15×10^{-1}	2.90×10^{-1}	2.76×10^{-1}
	TDMSER-TEQ	3.02×10^{-1}	2.70×10^{-1}	2.60×10^{-1}
	FD-MMSE-TEQ	6.48×10^{-2}	4.13×10^{-2}	3.43×10^{-2}
	DD-MMSE-TEQ	6.60×10^{-2}	2.42×10^{-2}	1.46×10^{-2}
4	TDDA-TEQ	1.09×10^{-1}	5.62×10^{-2}	3.84×10^{-2}
	TDMSER-TEQ	9.64×10^{-2}	4.88×10^{-2}	3.54×10^{-2}
	FD-MMSE-TEQ	1.19×10^{-2}	2.03×10^{-3}	1.62×10^{-3}
	DD-MMSE-TEQ	1.11×10^{-2}	7.10×10^{-4}	3.54×10^{-4}
6	TDDA-TEQ	2.82×10^{-2}	8.10×10^{-3}	4.60×10^{-3}
	TDMSER-TEQ	1.86×10^{-2}	3.30×10^{-3}	2.40×10^{-3}
	FD-MMSE-TEQ	3.63×10^{-3}	3.47×10^{-4}	2.60×10^{-4}
	DD-MMSE-TEQ	3.34×10^{-3}	2.02×10^{-4}	6.51×10^{-4}
8	TDDA-TEQ	1.38×10^{-2}	2.40×10^{-3}	1.30×10^{-3}
	TDMSER-TEQ	6.40×10^{-3}	8.00×10^{-4}	4.00×10^{-4}
	FD-MMSE-TEQ	2.68×10^{-3}	1.95×10^{-4}	1.66×10^{-4}
	DD-MMSE-TEQ	2.49×10^{-3}	6.51×10^{-5}	6.51×10^{-5}

the multipath effect remained significant, with a delay spread of approximately 20 ms, or about 120 symbol periods. During the DJKE1R1 and DJKG1R2 experiments, where the transmitter was mobile, the Doppler effect was significant. Both Doppler shifts were approximately 2.5 Hz. The multipath spread was about 25 ms (approximately 150 symbol periods) and 30 ms (approximately 180 symbol periods), respectively. The long delay spread and severe Doppler effect resulted in substantial distortion of the received signals.

B. Bit Error Rate Performance

The BER performance of various methods is presented below to demonstrate the effectiveness and superiority of the proposed method. The comparison methods include two time-domain adaptive turbo equalizers (TDDA-TEQ [14] and TDMSER-TEQ [7]) and a frequency-domain turbo equalizer (FD-MMSE-TEQ [19]). Table II indicates the experimental results of DJKC1R2 in terms of BER, illustrating the average BER performance of various receiving channels. Owing to the stationarity of the transceiver during the DJKC1R2 experiment, all methods demonstrate satisfactory detection performance. The TDDA-TEQ and TDMSER-TEQ perform inferiorly, particularly under constrained spatial diversity scenarios. As the number of channels increases, the performance of TDDA-TEQ and TDMSER-TEQ is improved, yielding a BER of less than 10^{-3} with 8-channel reception after three iterations. Equal-gain combining is used to extract spatial diversity gain. For FD-MMSE-TEQ and DD-MMSE-TEQ, superior detection performance can be attained even if constrained spatial diversity scenarios. Especially noteworthy is the fact that only DD-MMSE-TEQ achieves a BER below 10^{-1} in single-channel detection. The incorporation

of spatial diversity significantly enhances the performance of both methods. Following two iterations, the BER of FD-MMSE-TEQ and DD-MMSE-TEQ in a 6-channel reception scenario is less than 10^{-3} . Both methods exhibit the capability to achieve nearly error-free transmission in an 8-channel reception scenario. The proposed method requires only two iterations to attain a commendable BER, specifically less than 10^{-4} , with only a minimal number of incorrect bits. The experimental results of DJKC1R2 substantiate the effectiveness and superiority of the proposed method within slowly time-varying channels.

Two sets of mobile experiment results are presented as corroborative evidence, further certifying the performance of the proposed method in rapidly time-varying channels. The experimental results of DJKE1R1 and DJKG1R2 are illustrated in Tables III and IV, respectively. According to the results in Table III, the proposed method outperforms other methods in terms of BER with different receiving channels. During single-channel detection, only the proposed method achieves a BER below 10^{-1} after three iterations, whereas other methods failed to transmit in the intricate channel. With 2-channel detection, FD-MMSE-TEQ and DD-MMSE-TEQ demonstrate relatively acceptable performance, whereas TDDA-TEQ and TDMSER-TEQ prove ineffective. In scenarios with 6 and 8-channel detection, the DD-MMSE-TEQ achieves approximately error-free transmission. The proposed method has satisfactory performance, followed by FD-MMSE-TEQ. TDDA-TEQ and TDMSER-TEQ have the worst performance due to the slow convergence of the filters.

The pronounced Doppler effect significantly curtails communication reliability in experiment DJKG1R2. All methods exhibit unreliability when using only single-channel detection. Fortunately, as receiving channels increase, the proposed method has a BER of less than 10^{-1} (2-channel detection). With

TABLE III
AVERAGE BER PERFORMANCE OF EXPERIMENT DJKE1R1 WITH DIFFERENT METHODS

Channel Number	Method	Iter=1	Iter=2	Iter=3
1	TDDA-TEQ	4.57×10^{-1}	4.61×10^{-1}	4.58×10^{-1}
	TDMSER-TEQ	4.49×10^{-1}	4.44×10^{-1}	4.45×10^{-1}
	FD-MMSE-TEQ	1.68×10^{-1}	1.41×10^{-1}	1.37×10^{-1}
	DD-MMSE-TEQ	1.64×10^{-1}	1.03×10^{-1}	7.88×10^{-2}
2	TDDA-TEQ	3.69×10^{-1}	3.59×10^{-1}	3.53×10^{-1}
	TDMSER-TEQ	3.48×10^{-1}	3.20×10^{-1}	3.05×10^{-1}
	FD-MMSE-TEQ	8.73×10^{-2}	5.50×10^{-2}	5.08×10^{-2}
	DD-MMSE-TEQ	8.17×10^{-2}	1.71×10^{-2}	4.89×10^{-3}
4	TDDA-TEQ	1.54×10^{-1}	9.80×10^{-2}	7.58×10^{-2}
	TDMSER-TEQ	1.53×10^{-1}	9.54×10^{-2}	7.03×10^{-2}
	FD-MMSE-TEQ	1.79×10^{-2}	7.25×10^{-3}	7.23×10^{-3}
	DD-MMSE-TEQ	1.61×10^{-2}	1.16×10^{-3}	5.79×10^{-4}
6	TDDA-TEQ	1.17×10^{-1}	6.85×10^{-2}	4.98×10^{-2}
	TDMSER-TEQ	1.12×10^{-1}	6.16×10^{-2}	4.33×10^{-2}
	FD-MMSE-TEQ	1.29×10^{-2}	5.90×10^{-3}	5.69×10^{-3}
	DD-MMSE-TEQ	1.12×10^{-2}	1.01×10^{-3}	1.45×10^{-4}
8	TDDA-TEQ	1.14×10^{-1}	6.75×10^{-2}	5.07×10^{-2}
	TDMSER-TEQ	1.11×10^{-1}	5.84×10^{-2}	4.14×10^{-2}
	FD-MMSE-TEQ	1.35×10^{-2}	6.58×10^{-3}	6.55×10^{-3}
	DD-MMSE-TEQ	1.10×10^{-2}	1.01×10^{-3}	1.45×10^{-4}

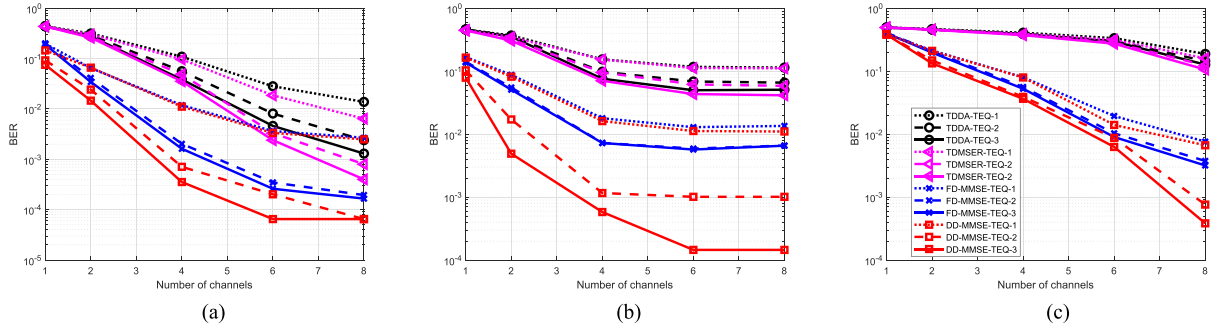


Fig. 12. Comparison of the average BER under different channel numbers. (a) DJKC1R2. (b) DJKE1R1. (c) DJKG1R2.

8-channel detection, only the BER of the proposed method draws near the 10^{-4} threshold, representing a one-order-of-magnitude improvement compared to FD-MMSE-TEQ. Nevertheless, even under these conditions, the BER of TDDA-TEQ and TDMSER-TEQ remains relatively high at 10^{-1} .

The results of Tables II–IV are more intuitively shown in Fig. 12. Both the experimental results of mobile and static communication substantiate the effectiveness and superiority of the proposed method. The FD-MMSE-TEQ performs the second, while time-domain methods are the worst. The proposed method demonstrates commendable effectiveness in both slowly and rapidly time-varying channels.

For DJKC1R2, DIKE1R1, and DJKG1R2, the frame BER results are illustrated in Figs. 13–15, respectively. They show the percentage of frames within specified BER ranges. The TDDA-TEQ, TDMSER-TEQ, FD-MMSE-TEQ, and DD-MMSE-TEQ are denoted as I to IV, respectively. For DJKC1R2, utilizing TDDA-TEQ and TDMSER-TEQ with 4-channel detection,

more than 50% of frames failed to be transmitted. Following three iterations, FD-MMSE-TEQ achieves error-free transmission for only 7% of frames. In contrast, the proposed method achieves correct decoding for 33% and 40% of frames following 2 and 3 iterations, respectively. With 8-channel detection, 87% of the frames are reliably transmitted after two iterations, and the BER for all frames is below 10^{-3} . Despite the enhanced performance of the comparative methods attributable to diversity gain, they remain incomparable to DD-MMSE-TEQ.

For DJKE1R1, utilizing TDDA-TEQ and TDMSER-TEQ with 4-channel detection, more than 80% and 60% of frames, respectively, failed to achieve a BER less than 10^{-1} in the first iteration. Conversely, with the proposed method, the BER of all frames is below 10^{-1} . As the number of iterations increases, the proposed method demonstrates the capacity to transmit more than 60% of frames without errors. With 8-channel detection, only DD-MMSE-TEQ has more than 80% of frames decoded correctly after three iterations.

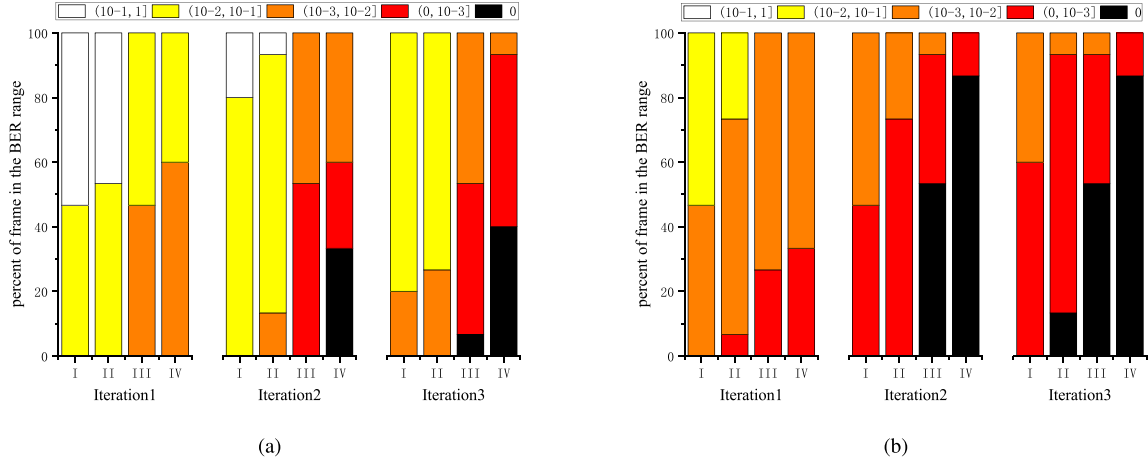


Fig. 13. Results of experiment DJKC1R2. We considered four channel equalization methods. I:TDDA-TEQ, II:TDMSE-TEQ, III:FD-MMSE-TEQ, IV:DD-MMSE-TEQ. (a) Channel=4. (b) Channel=8.

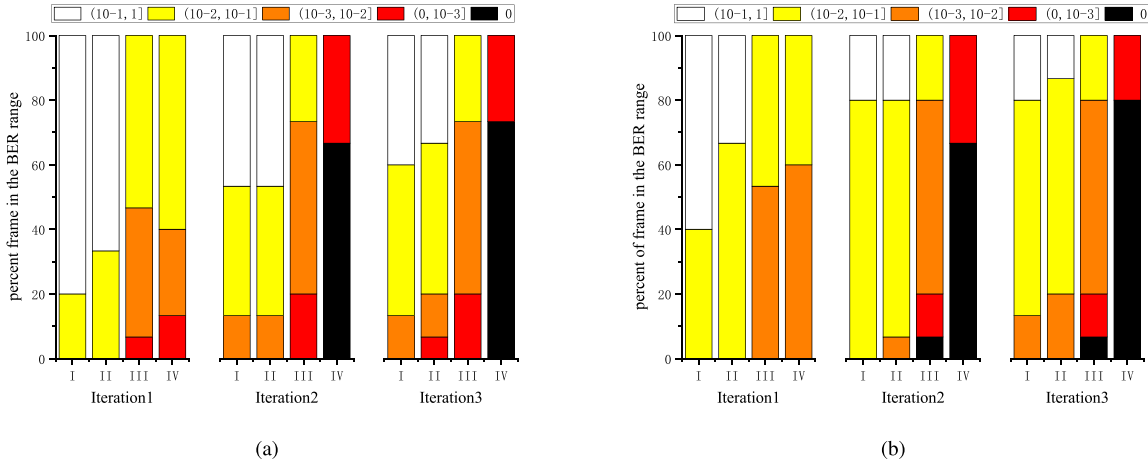


Fig. 14. Results of experiment DJKE1R1. We considered four channel equalization methods. I:TDDA-TEQ, II:TDMSE-TEQ, III:FD-MMSE-TEQ, IV:DD-MMSE-TEQ. (a) Channel=4. (b) Channel=8.

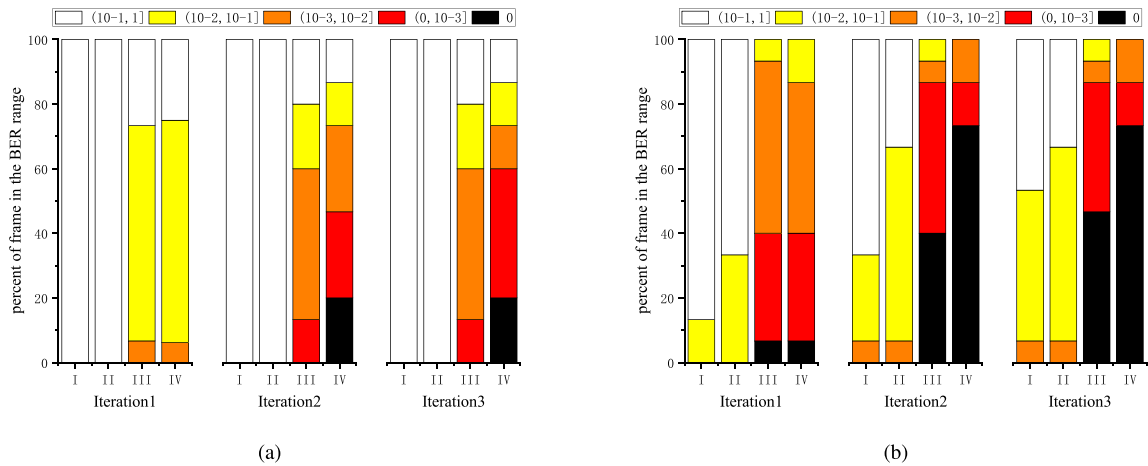


Fig. 15. Results of experiment DJKG1R2. We considered four channel equalization methods. I:TDDA-TEQ, II:TDMSE-TEQ, III:FD-MMSE-TEQ, IV:DD-MMSE-TEQ. (a) Channel=4. (b) Channel=8.

TABLE IV
AVERAGE BER PERFORMANCE OF EXPERIMENT DJKG1R2 WITH DIFFERENT METHODS

Channel Number	Method	Iter=1	Iter=2	Iter=3
1	TDDA-TEQ	4.95×10^{-1}	4.92×10^{-1}	4.92×10^{-1}
	TDMSER-TEQ	4.95×10^{-1}	4.92×10^{-1}	4.93×10^{-1}
	FD-MMSE-TEQ	3.89×10^{-1}	4.07×10^{-1}	4.01×10^{-1}
	DD-MMSE-TEQ	3.89×10^{-1}	3.82×10^{-1}	3.81×10^{-1}
2	TDDA-TEQ	4.66×10^{-1}	4.58×10^{-1}	4.59×10^{-1}
	TDMSER-TEQ	4.59×10^{-1}	4.56×10^{-1}	4.54×10^{-1}
	FD-MMSE-TEQ	2.13×10^{-1}	2.07×10^{-1}	1.97×10^{-1}
	DD-MMSE-TEQ	2.10×10^{-1}	1.48×10^{-1}	1.33×10^{-1}
4	TDDA-TEQ	4.08×10^{-1}	3.92×10^{-1}	3.88×10^{-1}
	TDMSER-TEQ	3.91×10^{-1}	3.81×10^{-1}	3.73×10^{-1}
	FD-MMSE-TEQ	7.98×10^{-2}	5.60×10^{-2}	5.26×10^{-2}
	DD-MMSE-TEQ	7.94×10^{-2}	3.91×10^{-2}	3.64×10^{-2}
6	TDDA-TEQ	3.34×10^{-1}	3.01×10^{-1}	2.96×10^{-1}
	TDMSER-TEQ	3.15×10^{-1}	2.83×10^{-1}	2.73×10^{-1}
	FD-MMSE-TEQ	1.94×10^{-2}	1.03×10^{-2}	8.99×10^{-3}
	DD-MMSE-TEQ	1.40×10^{-2}	8.72×10^{-3}	6.26×10^{-3}
8	TDDA-TEQ	1.88×10^{-1}	1.45×10^{-1}	1.27×10^{-1}
	TDMSER-TEQ	1.60×10^{-1}	1.23×10^{-1}	1.06×10^{-1}
	FD-MMSE-TEQ	7.60×10^{-3}	3.75×10^{-3}	3.18×10^{-3}
	DD-MMSE-TEQ	6.65×10^{-3}	7.60×10^{-4}	3.83×10^{-4}

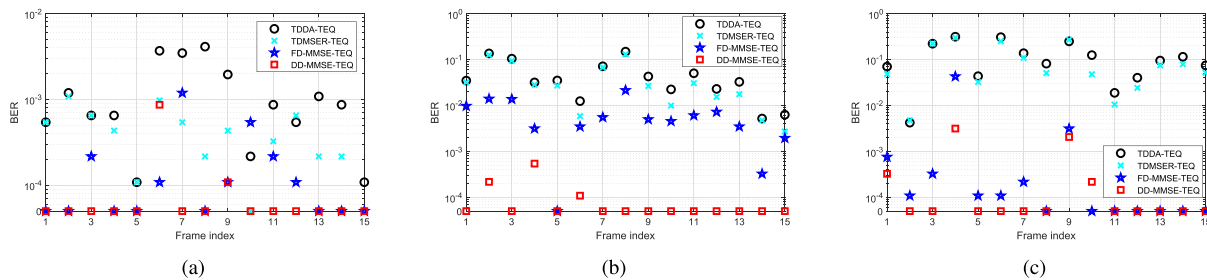


Fig. 16. Average BER of each frame with eight channels. (a) DJK1R2. (b) DJKE1R1. (c) DJKG1R2.

For DJKG1R2, when using 4-channel detection, all data frames detected by TDDA-TEQ and TDMSER-TEQ have a BER of more than 10^{-1} after three iterations. However, employing FD-MMSE-TEQ and DD-MMSE-TEQ resulted in all frames attaining a BER below 10^{-2} , indicating enhanced robustness in channel equalization. Despite using diversity gain, time-domain methods still perform poorly. While the performance of FD-MMSE-TEQ has significantly improved, it remains inferior to DD-MMSE-TEQ, which achieves a notable 73% of frames transmitted without errors. To more intuitively illustrate the performance differences between different methods, the BER of each frame is shown in Fig. 16. It is obvious that the proposed method enables error-free transmission of the most frames.

Fig. 17 shows constellation diagrams of the equalized symbols utilizing TDDA-TEQ, TDMSER-TEQ, FD-MMSE-TEQ, and DD-MMSE-TEQ. Due to space constraints, only the results of the initial and third iterations are provided. The first and second

columns correspond to the results of the first and third iterations of DJK1R2, respectively, and the other methods follow a similar pattern. Following the DD-MMSE-TEQ equalization, the equalized symbols exhibit the highest degree of concentration, resulting in the highest output SNR. In contrast, constellations from alternative methods appear dispersed, spanning the entire plane. Through iterative equalization, the equalized symbols converge around the ideal constellation points, thereby further enhancing performance. The soft-decision symbols converge to the ideal constellation points during the iterations, as illustrated in Fig. 18. The DD-MMSE-TEQ can converge to the QPSK symbol position after three iterations, in sharp contrast to the comparison methods, which show more divergent constellation points. From the point of view of constellations, the soft decision symbols are more reliable than equalized symbols due to the use of soft information. External information is generated by the soft decoder and then fed back into prior information for subsequent iterations. In subsequent iterations, these more reliable soft

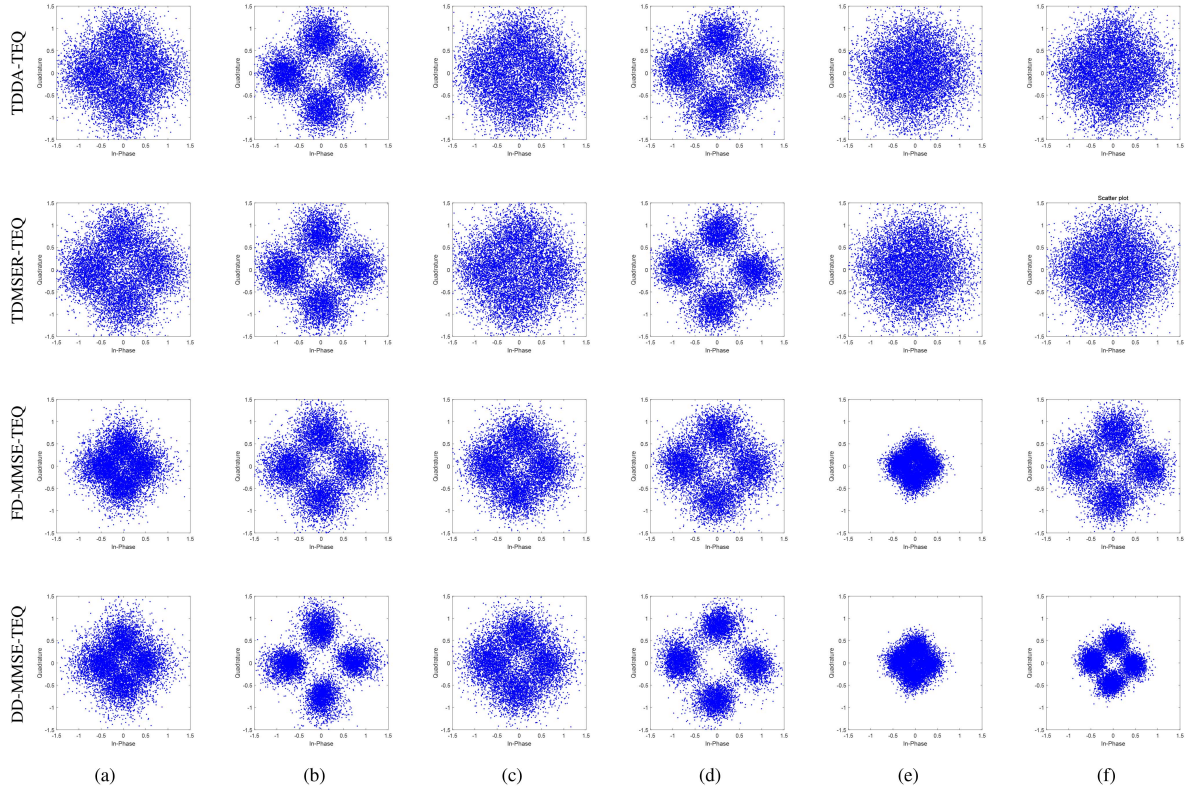


Fig. 17. Constellation diagrams of the equalized symbols. Three iterations are conducted using TDDA-TEQ, TDMSE-TEQ, FD-MMSE-TEQ, and DD-MMSE-TEQ. (a) C1R2-I1. (b) C1R2-I3. (c) E1R1-I1. (d) E1R1-I3. (e) G1R2-I1. (f) G1R2-I3.

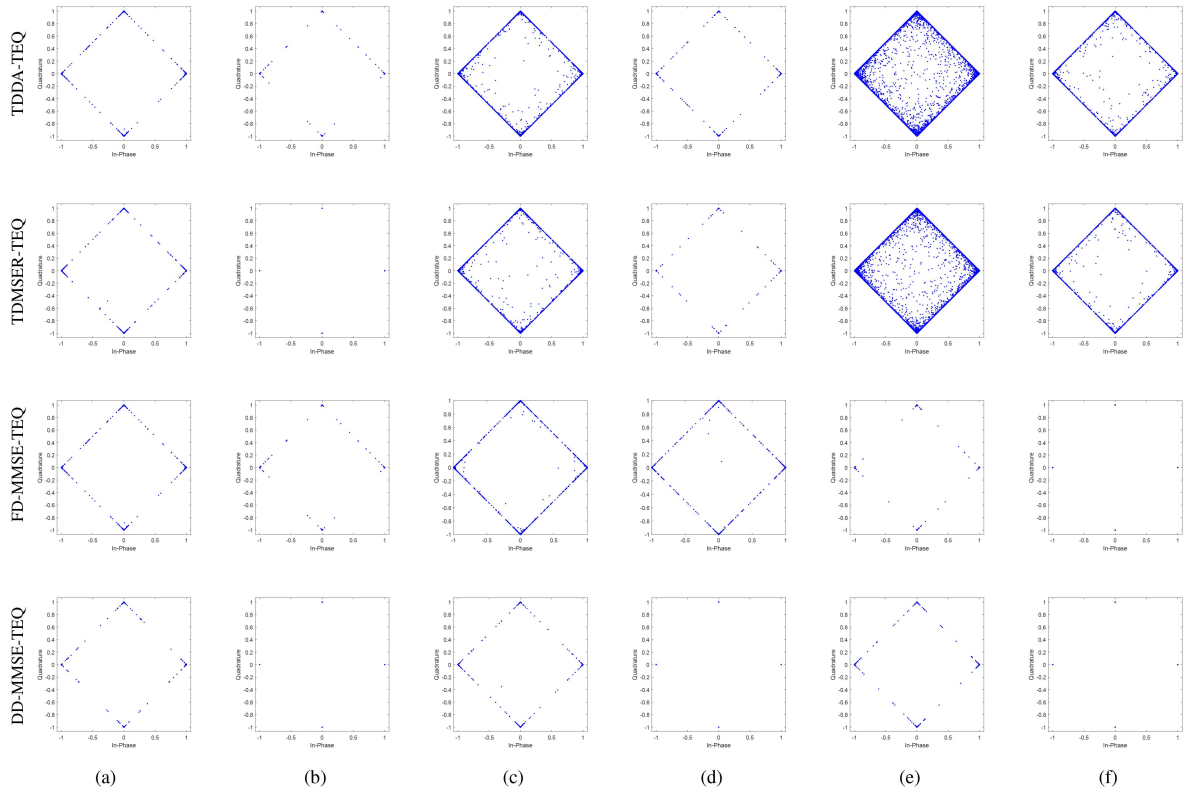


Fig. 18. Constellation diagrams of the soft-decision symbols. Three iterations are conducted using TDDA-TEQ, TDMSE-TEQ, FD-MMSE-TEQ, and DD-MMSE-TEQ. (a) C1R2-I1. (b) C1R2-I3. (c) E1R1-I1. (d) E1R1-I3. (e) G1R2-I1. (f) G1R2-I3.

TABLE V
COMPUTATIONAL COMPLEXITY OF THE CHANNEL EQUALIZATION

Method	Complexity
TDDA-TEQ	$3I_{max}K(L_f + L_b + 1)$
TDMSER-TEQ	$3I_{max}K(L_f + L_b + 5)$
FD-MMSE-TEQ	$I_{max}(3K(1 + 1/2\log_2 K))$
DD-MMSE-TEQ	$I_{max}K^3$
OTFS-MMSE-TEQ	$I_{max}K^3$

decision symbols will be reintroduced to guide the equalizer for interference cancellation, consequently yielding exceptional performance.

To briefly summarize the previous analysis, the following observations can be made. First, compared to other methods, DD-MMSE-TEQ exhibits the most favorable BER performance. DD-MMSE-TEQ transforms the rapidly time-varying channels in the TF-domain into approximately time-invariant channels through the DD domain transformation. It has the ability to improve the challenge posed by rapidly time-varying channels. As a result, the performance achieved by DD-MMSE-TEQ is not only satisfactory but also exceeds that of other comparison methods. Moreover, the performance of FD-MMSE-TEQ is partially degraded in rapidly time-varying channels because the quasi-static assumption is no longer satisfied. The performance of TDDA-TEQ and TDMSER-TEQ is not satisfactory. The insufficient training sequence and the rapidly time-varying channels result in the filter failing to converge. When decision-directed equalization is used, it causes significant error propagation, thereby degrading the performance of TDDA-TEQ and TDMSER-TEQ. Ultimately, DD-MMSE-TEQ shows enhanced iterative gain compared to other methods. During the initial iteration, the performance enhancement of DD-MMSE-TEQ over FD-MMSE-TEQ is comparatively marginal. The performance of DD-MMSE-TEQ is improved in successive iterations due to the transfer of soft information between the DD domain and time-domain through unitary transformation. The cross-domain processing markedly diminishes the probability of error propagation, rendering the iterative gain more pronounced compared to other comparison methods.

VI. COMPUTATIONAL COMPLEXITY

The computational complexity of all methods involved in the simulation and experiment is analyzed and listed in Table V. Only the number of complex multiplications required by the equalizer is considered. I_{max} represents the maximum number of iterations. The lengths of the feedforward filter and feedback filter in TDDA-TEQ and TDMSER-TEQ are indicated by L_f and L_b , respectively. According to (36), the complexity of DD-MMSE-TEQ mainly arises from matrix inversion. The complexity of DD-MMSE-TEQ is approximately on the order of $O(K^3)$. The channel matrix of FD-MMSE-TEQ is a diagonal matrix, and the complexity of the feedforward filter and feedback filter is reduced from $O(K^3)$ to $O(K)$. Finally, the overall complexity of FD-MMSE-TEQ is $O(3K(1 + 1/2\log_2 K))$ [19]. The complexity of TDDA-TEQ and TDMSER-TEQ mainly arises

from the processes of filter updating and adaptive filtering. In terms of the required multiplications, the complexity of TDDA-TEQ and TDMSER-TEQ is almost identical. However, TDMSER-TEQ involves more add operations and additional tanh calculations [8]. The method proposed has a higher complexity, but its performance is improved compared to other methods. This article presents a preliminary idea about single-carrier DD domain equalization, so the low-complexity processing method has not been considered. Reducing the computational overhead of single-carrier DD domain equalization remains an open and urgent problem.

VII. CONCLUSION

In this article, we propose a novel DD domain turbo equalizer for single-carrier UWA communications that mitigates the performance degradation commonly encountered in rapidly time-varying UWA channels. The proposed method combines symbol estimation in the DD domain, symbol detection in the time-domain and soft information interaction in the cross-domain. The equalizer operating in the DD domain transforms the rapidly time-varying channels into quasi-static channels, enhancing the system's adaptability to time dynamics. Soft information interaction between the time-domain and DD domain is realized through unitary transformation to enhance iterative gain. This cross-domain information exchange allows the errors that often persist in the time-domain to be effectively dispersed throughout the DD domain. This, in turn, dramatically reduces the possibility of error propagation and improves detection reliability. Simulation results indicate that the proposed DD-MMSE-TEQ outperforms FD-MMSE-TEQ in terms of bit error rate and achieves a performance level comparable to conventional OTFS, albeit at the cost of increased computational complexity at the receiver. The proposed method has been validated by the field data collected in a mobile UWA communications experiment conducted in Danjiangkou Reservoir, Henan Province, China, in July 2022.

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